Predicting Alzheimer's Disease Stages from Microbiome Profiles Using Genome-Scale Metabolic Models and Machine Learning

1. PROJECT OVERVIEW AND OBJECTIVES

1.1 Problem Statement

This project aims to build a computational data-driven diagnostic framework for Alzheimer's disease (AD) by combining microbiome-derived genome-scale metabolic models (GEMs) with machine learning. The main goal is to predict the disease stage from gut microbial activity, allowing us to move towards a non-invasive and scalable approach for diagnosing AD.

1.2 Existing Solutions and Gaps

Alzheimer's disease is a progressive neurodegenerative disorder that remains the leading cause of dementia worldwide, with over 50 million individuals estimated to be affected and no available cure [1]. Current diagnostic practices rely on a combination of cerebrospinal fluid (CSF) and plasma biomarkers as well as neuroimaging techniques such as positron emission tomography (PET) and magnetic resonance imaging (MRI) [2]. However these current methods face significant limitations. PET imaging is costly and thus inaccessible to most clinics, whereas MRI alone cannot reliably confirm AD pathology [1]. Likewise, CSF and plasma biomarkers, such as amyloid-β (Aβ) and phosphorylated tau, lack definitive predictive power on their own, thus making ascertain diagnosis only possible post mortem [1,2]. Furthermore, CSF collection requires a lumbar puncture procedure which is not only invasive but also linked to adverse side effects such as headaches and back pain, limiting its use in clinical settings [3]. Collectively these current limitations highlight the need for a diagnostic method of AD that is not only accessible, non-invasive, and cost effective, but also accurate and scalable for AD diagnosis in all clinical settings.

In recent years, the microbiota-gut-brain axis has emerged as a key area of investigation, connecting gut microbial metabolism and neurodegenerative processes in AD [4]. This bidirectional network connects the gut and central nervous system through neural, immune, and metabolic signalling pathways [4]. The gut microbiota produces metabolites such as short chain fatty acids (SCFAs), neurotransmitters, and immune mediators that can modulate neuroinflammation and brain function [5]. Disruption in microbial composition is called gut dysbiosis and it has been shown to increase gut and blood-brain barrier permeability, elevate inflammatory signalling, and alter amyloid and tau metabolism, all of which collectively contribute to AD progression [5]. Therefore the gut microbiome may be the key to providing a measurable, non-invasive biomarker for early detection of AD. These microbial and metabolic alterations can be quantified through 16S rRNA and metagenomic sequencing, paired with metabolomic profiling to

produce quantitative data on microbial composition and metabolite changes associated with AD [4,5]. Engineering and computational tools such as data driven modelling provides us the ability to integrate these biological markers into predictive frameworks capable of diagnosing AD. Thus, a biomedical engineering approach is needed to translate microbiome data into a reproducible and scalable diagnostic system for AD detection.

Current research demonstrates the importance of integrating microbiome profiles with metabolic data to better understand the mechanisms underlying neurodegenerative diseases [6]. A recent study used 16S rRNA sequencing in Alzheimer's disease mouse models and combined microbiome data with brain metabolomics to identify shared pathways linking microbial composition to neuroinflammation [6]. They found that changes in glycerophospholipid metabolism were associated with both microbial shifts and immune-related brain inflammation [6]. Although this study relied on experimental rather than computational integration, it provides clear support for coupling 16S rRNA data with metabolic modeling frameworks such as GEMs, or metagenomic-scale modelling (MICOM) to predict how microbial community composition influences metabolic activity and neurodegenerative disease outcomes [6].

Building on this, other studies have highlighted that while single-species GEMs are well established, the integration of multi-omics metadata within community-level GEMs remains in its early stages and requires further computational innovation [7]. Recent advances in constraint-based and machine-learning-assisted modeling have been applied to improve the predictive accuracy and reduce uncertainty in GEMs [7]. These developments underscore the growing importance of combining microbiome sequencing, metabolic modeling, and data-driven analysis to capture complex microbial interactions . This aligns with the direction of the present work, which aims to integrate 16S-derived community composition data with metabolic simulation frameworks to explore microbiome-driven metabolic processes along the gut–brain axis.

Existing literature shows that gut microbiome composition can influence brain microglia as a driver for neurodegenerative diseases [8]. A study that explored machine learning as a means to determine the incidence of AD based on population gut microbiome profiles used a machine learning pipeline built in R using the "Statistical Inference of Associations between Microbial Composition And host phenoTypes" (SIAMCAT) package [9]. They used two large 16S rRNA datasets and then performed a number of statistical analyses such as alpha and beta diversity analyses, and permutational multivariate analysis of variance (PERMANOVA) tests [9]. The alpha diversity was measured as the diversity within the microbiome population to find the number and distribution of species present [9]. The study then used the Wilcoxon signed-rank test at

an alpha level of 0.05 to compare alpha diversity between countries with a high vs low prevalence of AD [9]. Their beta diversity test was used to measure the diversity between the microbiome populations and used the Bray–Curtis distance matrix, which quantifies dissimilarity between samples based on relative microbial abundance [9]. To test for significant compositional differences, a PERMANOVA test was used to determine whether between group variation exceeded within group variation [9]. These statistical analyses served as a critical preprocessing step as they prove that the microbiome data actually contained a meaningful, disease-related signal for the model to learn from [9].

This study then utilized the SIAMCAT machine learning model to predict AD patterns and cross-validated using a 10× resampled 10-fold test [9]. In order to test the models ability to distinguish between data groups, receiver operating characteristic (ROC) and precision recall curves were used, which provided insight as to the accuracy and consistency of the model to predict patterns in the gut microbiome linked to AD [9]. This study did not examine direct changes in the brain and therefore, did not measure brain structure, function, or pathology [9]. Overall findings concluded that measurable and significant differences existed in the microbial data, which the SIAMCAT machine learning pipeline then leveraged to identify specific taxa patterns [9]. Studies like this demonstrate that measurable and disease specific patterns exist in gut microbial composition and can be identified using machine learning. Building on this foundation, our work extends beyond microbial composition differences by applying machine learning to predict AD stages based on the metabolic interactions that drive these patterns.

Despite existing research linking gut microbiota and Alzheimer's disease, most studies remain correlation-based and focus more on taxonomic shifts rather than the metabolic interactions that lead to neurodegeneration [10]. The use of genome-scale metabolic models and community modeling tools like MICOM provides a valuable method to predict how microbes interact metabolically. However, their application in Alzheimer’s disease microbiome research is still limited and rare. There are few validated GEMs designed to AD-associated microbial taxa, predictions of metabolite fluxes can be inconsistent and uncertain [11]. Furthermore, the integration of data across different biological scales, such as genomic, transcriptomic, and metabolomic levels, is often inadequate, leading to oversimplified models that do not accurately represent the complexity of AD pathophysiology [12]. Reproducibility presents a major challenge in microbiome research. Many studies on microbiomes and AD use small, geographically restricted groups and different sequencing methods. Additionally, publicly available AD microbiome datasets vary in the completeness of their metadata, making it hard to validate findings across studies[13]. Additionally, machine learning approaches also

tend to rely solely on relative abundance data without incorporating metabolic flux simulations and limits their interpretability and predictive capabilities [14].
These gaps demonstrate the need for a data-driven framework that combines microbiome-derived metabolic modeling and advanced predictive analytics to better understand how Alzheimer's disease develops and to make stage predictions more accurate.

1.3 Project Objectives

The objective of this project is to develop and evaluate a computational diagnostic framework that integrates microbiome taxonomic data with genome-scale metabolic modeling and machine learning to predict cognitive disease stages (Healthy, Mild Cognitive Impairment, and Alzheimer's disease).

Specifically, this study aims to:

1. Extract and preprocess genus-level relative abundance features from publicly available 16S rRNA sequencing datasets.
2. Develop community genome-scale metabolic models using MICOM to derive functional metabolic flux features representing microbial biochemical activity.
3. Train and evaluate machine learning classifiers (Support Vector Machines and Random Forest models) using:
    - Taxonomic features alone.
    - Taxonomic features and metabolic features.
4. Compare model performance using accuracy, weighted precision, recall, F1 score, and confusion matrix analyses to assess:
    - Generalization performance
    - Class-specific detection capability
    - Impact of functional metabolic feature integration.

The overarching goal is to determine whether functional metabolic modeling enhances disease classification performance beyond taxonomic profiling alone, thereby supporting development of scalable and non-invasive tools for early Alzheimer's disease detection.

2. METHODOLOGY

This study integrates taxonomic profiling, microbial community metabolic modeling, and machine learning classification to investigate microbial differences across Healthy, Mild Cognitive Impairment (MCI), and Alzheimer's Disease (AD) samples. Our methodological pipeline consists of three stages:

1. Taxonomic preprocessing used for relative abundance extraction
2. Genome-scale metabolic modelling using MICOM

3. Machine learning classification using combined taxonomic and metabolic features.

2.1 Dataset 1: Taxonomic Processing and Relative Abundance (RA) Feature Extraction

Raw 16s rRNA sequencing data was obtained from NCBI PRJNA496408.
The metadata for this dataset included cohort information such as age, sex, BMI, and comorbidities. These features were imported into a pandas DataFrame. To ensure relevance to our analysis, human and mouse samples were first separated into distinct DataFrames, and only the human samples were retained. The human cohort was then divided into Alzheimer's disease (AD), mild cognitive impairment (MCI) and healthy control groups.

Sample-specific SRR identifiers were used with the SRA toolkit to find the paired-end FASTQ files for each of the 93 samples. Paired-end FASTQ files associated with each SRR ID were downloaded and stored in a structured dictionary. Across the 93 samples, taxonomic profiling identified 84 genera in AD, 95 in MCI and 87 in healthy samples. Ten genera were unique to MCI, seven were unique to AD, and three were unique to the healthy samples. These differences reflect the heterogeneous nature of the gut microbiome and support the need for feature filtering to reduce the sparsity in downstream processing. All downstream processing was conducted using QIIME2. Paired-end reads were imported and denoised using the DADA2 algorithm, which performs trimming, filtering, denoising, and subsequent merging of paired reads to find amplicon sequence variants (ASV).

Taxonomic assignment of the ASV was performed using SILVA138 Naive Bayes weighted for human-stool. Although the ASV were collapsed to the species level, no unique species level assignments could be found, which is a known limitation of SILVA138. Therefore, genus level classification served as the highest reliable taxonomic classification for our study. QIIME2 was used to generate genus-level RA tables for all samples. These RA tables were exported to pandas as CSV files for further processing. Samples with incomplete genus identification metadata were removed. These steps were repeated for each of the AD, MCI and healthy conditions.

Feature engineering was then performed to find the interpretable inputs for the machine learning models. These methods were conducted as follows:

1. Prevalence filtering: Genera present in less than 10% of the samples per were removed to reduce sparsity. This process was repeated per condition.
2. Differential RA features: The top five genera showing the greatest RA differences across Health, MCI, and AD were extracted and used as features for the

machine learning models.To find the genera with the most associated disease states, we calculated the range of the relative abundance as maximum minus minimum across AD, MCI and healthy means. This method was used as an alternative to statistical variance and captures genera whose abundance differs most strongly between conditions.

3. Dimensionality reduction: RA matrices were standardized using scikit-learn StandardScaler, followed by Principal Component Analysis (PCA) to find the first three principal components representing major axes for microbial variation.

We did not use the most abundant genera as the features because the dominant taxa were highly similar across all the conditions, reflected by the common core human-gut microbiota. Since these genera do not vary meaningfully across disease states, they were excluded from feature selection. These preprocessing steps formed dataset 1 for machine learning analysis.

2.2 Dataset 2: Metabolic Modeling Using MICOM

To model community metabolism, genus-level RA tables were mapped to genome-scale metabolic models (GEMs) from the AGORA database. This database was filtered to include only GEMs corresponding to the genera detected in our dataset.The cobra toolbox was used to validate the GEM files and remove incomplete metabolic models. For each sample, microbial community models were built using MICOM's *build* method. Communities could only be built for 25 samples due to missing or low-abundance GEMs (including 11 AD, 7 MCI and 7 healthy). Samples lacking sufficient GEMs were excluded because communities would not grow or were dominated by genera from negligible RAs.

MICOM cooperative trade-off simulations were performed for each sample to find community growth rates and reaction level metabolic fluxes. Fluxes were analyzed so that positive values indicated nutrient uptake and negative values indicated metabolite secretion. Using pandas, the top five reactions showing the largest flux differences across healthy, MCI and AD were found.
These features formed dataset 2 for the machine learning analysis.

2.3 Machine Learning Experiments

All RA and MICOM derived features were imported and merged into pandas as TSV files. Disease condition labels were used as ground-truth targets.

Four models were evaluated using scikit-learn:

1. Support Vector Machine (SVM) using only RA features
2. Random Forest using only RA features

3. SVM using RA and MICOM features
4. Random Forest using RA and MICOM features

For the SVM and Random Forest using RA only features, 5-fold cross validation was used because the full dataset contained 93 samples, or around 30 per condition. This allowed for reliable stratified splits, Final evaluation was also done using a 70/30 train-test split.

For the SVM and Random Forest models relying on RA and MICOM features, cross-validation was not used as only 25 samples were available, which would leave about 2 samples per condition per fold, rendering cross-validation statistically invalid.

2.4 Justification of Machine Learning Methods

Micobiome RA and MICOM flux data are high-dimensional, sparse and non-linear. SVM performs well in high-dimensional spaces and can find optimal separating boundaries with limited samples. In comparison, Random Forest is not sensitive to noise, handles correlated features and reduces overfitting by aggregating many decision trees. Both methods work well on small and imbalanced datasets, making them suitable for our study.

The high-dimensionality of the data can be attributed to the RA containing around 80-95 genera per group, while the MICOM adds an extra 6 features through the flux reactions (accounting for 5 extra added features), growth (1 extra added feature), and RA PCA components. These 6 extra features are generated per sample. Flux distributions come from MICOM optimization and are non-linear outputs from the constraint based model. This makes the relationship between RA and conditions non-linear.

3. RESULTS

3.1 Feature Set 1: Relative Abundance and Principal Component Results

QIIME2 processing identified 84 genera in AD samples, 95 in MCI samples, and 87 in healthy samples after removing unclassified microbes. Further processing revealed genera exclusive to each disease state, with 7 unique to AD, 10 to MCI, and 3 to healthy samples.

The mean relative abundance (MRA) per genus was calculated for each condition subset, and the inter-condition range (ICR) was subsequently determined to identify genera that differed the most in MRA across AD, MCI, and healthy groups.

Table 1 summarizes the results for the five genera with the largest inter-condition range. *Escherichia* demonstrated the greatest disparity, with a ~14% increase in MRA in AD

compared to MCI. A substantial difference was also observed between the AD and healthy groups, with AD samples showing ~10% higher MRA than healthy samples. *Bacteroides* had similar MRA in AD and healthy samples (~7%), but exhibited a ~10% increase in the MCI group. *Blautia* displayed fairly similar MRA in AD and MCI groups (~7–8%), with a ~5% higher MRA in the healthy group. The remaining genera had comparatively smaller differences (<3%) in MRA across the three groups.

Table 1 Mean relative abundance of genera with largest inter-condition range

Genus	**Mean Relative Abundance (%)[1]**			**Inter-condition Range (%)[2]**
	AD	***MCI***	***Healthy***	
Escherichia	21.1	6.86	10.9	14.3
Bacteroides	7.27	16.6	7.08	9.57
Blautia	8.29	7.30	12.5	5.15
Alistipes	1.37	3.94	1.69	2.57
Limosilactobacillus	2.59	0.60	0.31	2.27

[1] Mean relative abundance per genus across all samples within the AD, MCI, or Healthy subsets.

[2] Inter-condition range calculated as the difference between the maximum and minimum mean relative abundance results across the AD, MCI, and healthy subsets.

Prevalence filtering was applied to the AD, MCI, and healthy subsets to retain genera present in at least a threshold percentage of samples in any condition. An initial threshold of 10% reduced the total number of genera to 62 in the combined relative abundance dataset.

Principal component analysis (PCA) was performed on the filtered dataset to characterize three axes of variation in genus-level relative abundance across all samples. The first principal component (PC1) explained ~8% of total variance, PC2 explained ~7%, and PC3 explained ~6%. Cumulatively, the first three components captured ~21.2% of total variance following 10% prevalence filtering.

Increasing the prevalence filtering threshold to 20% reduced the number of retained genera to 42 in the combined dataset. PCA performed on the 20% filtered dataset showed a slight increase in cumulative explained variance to ~25.2%. PC1 accounted for ~10% of total variance, PC2 explained ~8%, and PC3 explained 7%.

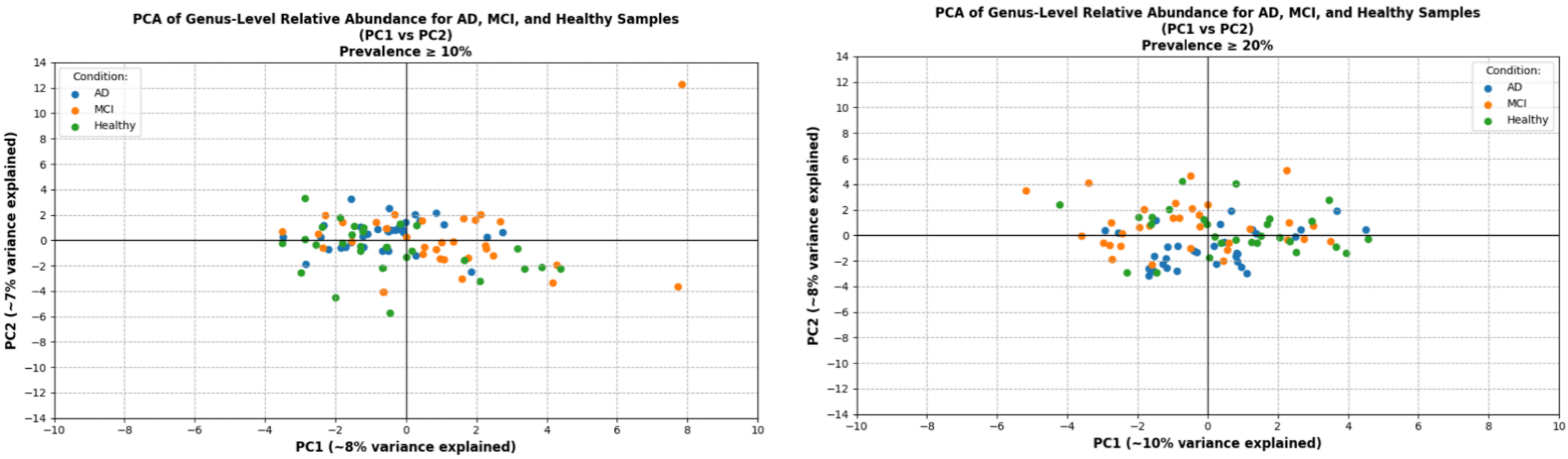

Figure 1 Principal component analysis results after 10% and 20% prevalence filtering. Each point corresponds to PC1 and PC2 scores for an individual sample, colored by condition group (AD, MCI, Healthy).

Figure 1 compares the PC1 vs. PC2 score plots for both prevalence thresholds. In both cases, the plots illustrate that principal component scores overlap across AD, MCI, and healthy samples. The dispersion of scores varies slightly between the two plots, with the 10% prevalence plot being slightly stretched due to outlying MCI scores. However, neither plot exhibits distinct clustering of condition groups along the first two principal components.

The final feature set for relative abundance contains the top five genera with the largest inter-condition range and the PC1-PC3 scores obtained after 10% prevalence filtering.

3.2 Feature Set 2: MICOM Metabolic Flux and Community Growth

MICOM reconstructed communities for 25 total samples, including 11 communities for AD, 7 for MCI, and 7 for healthy. Missing GEM models prevented community building for the remaining samples, resulting in a smaller MICOM feature set (n = 25) compared to the relative abundance feature set (n = 93).

Mean cooperative-tradeoff community growth rate was highest for AD samples at 53. Healthy and MCI samples had fairly similar mean community growth rates of 47 and 45, respectively.

Similar to the MRA calculation in the relative abundance analysis, the mean metabolic flux (MMF) per reaction was determined for each condition group, followed by computation of MMF inter-condition range. The five reactions that differed the most in MMF across AD, MCI, and healthy groups are summarized in Table 2.

Table 2 Mean metabolic flux of reactions with largest inter-condition range

Reaction[1]	**Mean Metabolic Flux**[2] **(mmol·gDW^{-1}· h^{-1})**			**Inter-condition Range**[3] **(mmol·gDW^{-1}· h^{-1})**
	AD	***MCI***	***Healthy***	
EX_xyluglc_m	-1.74	-0.456	-282	281
EX_rpn_96990_glc_m	2.70	0.753	250	248
EX_ac_m	443	204	421	239
EX_co2_m	343	144	360	216
EX_7ocholate_m	-199	-1.54	0.010	199

[1] EX_xyluglc_m: xyluglucan exchange; EX_rpn_96990_glc_m: glucose exchange; EX_ac_m: acetate exchange; EX_co2_m: carbon dioxide exchange; EX_7ocholate_m: 7-oxocholate exchange.

[2] Mean metabolic flux per reaction across all samples within the AD, MCI, or healthy subsets. Positive flux indicates uptake of metabolite and negative flux indicates secretion, expressed in units of millimoles per gram dry weight per hour (mmol·gDW^{-1}· h^{-1}).

[3] Inter-condition range calculated as the difference between the maximum and minimum mean metabolic flux results across the AD, MCI, and Healthy subsets.

The xyluglucan exchange reaction had the largest MMF-ICR (281 mmol·gDW^{-1}· h^{-1}), followed by the glucose (248 mmol·gDW^{-1}· h^{-1}), acetate (239 mmol·gDW^{-1}· h^{-1}), carbon dioxide (216 mmol·gDW^{-1}· h^{-1}), and 7-oxocholate (199 mmol·gDW^{-1}· h^{-1}) exchange reactions. The xyluglucan exchange reaction had increased secretion in the healthy communities compared to the AD and MCI groups, while the glucose exchange reaction had increased uptake in the healthy communities. Acetate and carbon dioxide exchange reactions had similar uptake in AD and healthy communities but reduced uptake in MCI samples. The 7-oxocholate exchange reaction had near-zero flux in MCI and healthy communities but exhibited increased secretion in AD communities.

3.3 Machine Learning Models: Model A

A Support Vector Machine (SVM) classifier with a radial basis function (RBF) kernel was trained using only RA features to assess whether taxonomic composition alone can distinguish among AD, MCI, and healthy controls. All input features were standardized prior to modelling. Performance was evaluated using both 5-fold cross-validation on the training dataset and single-test evaluation on an independent hold-out test set to assess real world generalization. 5-fold cross-validation produced a mean accuracy of 58.5%, however, overall testing accuracy declined to 36%.

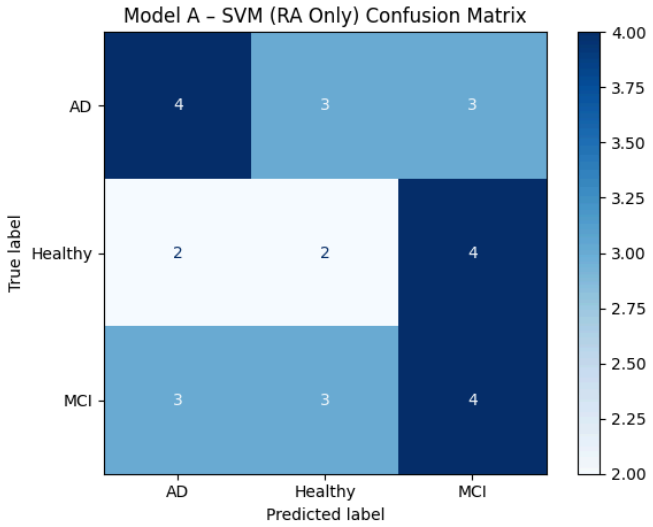

Figure 2 Confusion matrix for Model A (SVM, RA only). Predicted versus true class labels for Alzheimer's disease (AD), mild cognitive impairment (MCI), and healthy controls.

Figure 2 presents the confusion matrix for Model A. Classification errors occurred across all classes with no group showing consistently strong prediction performance. While subsets of AD and MCI samples were correctly identified a large proportion of cases were misclassified into alternative categories. Healthy controls were in particular an issue with frequent confusion across both disease groups.

Table 3 Classification metrics for Model A (SVM, RA only). Class-specific precision, recall, and F1-scores calculated on the held-out test set. Weighted averages summarize overall model performance.

	Precision	**Recall**	**F1-score**
AD	0.44	0.40	0.42
Healthy	0.25	0.25	0.25
MCI	0.36	0.40	0.38
Weighted Average	0.36	0.36	0.36

Class-specific precision, recall and F1-scores are summarized in Table 3. The AD group achieved 44% precision and 40% recall (F1 = 0.42). MCI predictions were similarly modest (precision = 36%, recall = 40%, F1 = 0.38). Performance was weakest for healthy controls (precision = 25%, recall = 25%, F1 = 0.25), showing high rates of false positives and false negatives. Weighted average (precision = 0.36, recall = 0.36, F1 =

0.36) matched the low overall test accuracy, confirming uniformly poor classification across all clinical categories.

3.4 Machine Learning Models: Model B

A Random Forest (RF) classifier was trained using only RA features to evaluate whether ensemble learning could improve disease classification performance relative to the SVM when relying solely on taxonomic data. The model consisted of 500 decision trees with balanced class weights to mitigate minor class imbalance. Performance was assessed using both 5-fold cross-validation on training data and independent evaluation on the test set. Cross-validation achieved a mean accuracy of 49.2%, but test-set accuracy declined to 32%.

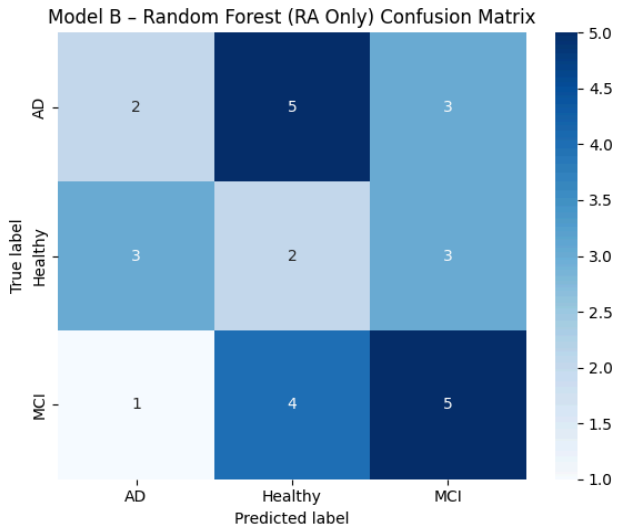

Figure 3 Confusion matrix for Model B (Random Forest, RA only). Predicted versus true labels for AD, MCI, and healthy participants.

Table 4 Classification metrics for Model B (Random Forest, RA only). Precision, recall and F1-scores calculated for each class on the held-out test dataset. Weighted averages summarize overall performance.

	Precision	**Recall**	**F1-score**
AD	0.33	0.20	0.25
Healthy	0.18	0.25	0.21
MCI	0.45	0.50	0.48
Weighted Average	0.33	0.32	0.32

Figure 3 displays the confusion matrix for Model B, and detailed metrics are summarized in Table 4. Prediction performance was strongest for the MCI group, where precision reached 45%, recall 50% and F1-score 0.48. In contrast, performance for Alzheimer's disease was poor (precision 33%, recall = 20%, F1 = 0.25) with most AD cases misclassified as either MCI or healthy participants. Healthy controls again yielded the weakest outcomes (precision = 18%, recall = 25%, F1 = 0.21), reflecting extensive overlap between healthy and disease microbiome compositions when evaluated using RA features alone. Weighted averages (precision = 0.33, recall = 0.32, F1 = 0.32) paralleled the low overall accuracy.

3.5 Machine Learning Models: Model C

A Support Vector Machine (SVM) classifier with a radial basis function (RBF) kernel was trained using both relative abundance (RA) taxonomic features and functional metabolic outputs derived from MICOM modeling. All input variables were standardized prior to training. Model performance was evaluated using single-test assessment on an independent held-out dataset to determine real world generalization following functional feature integration.

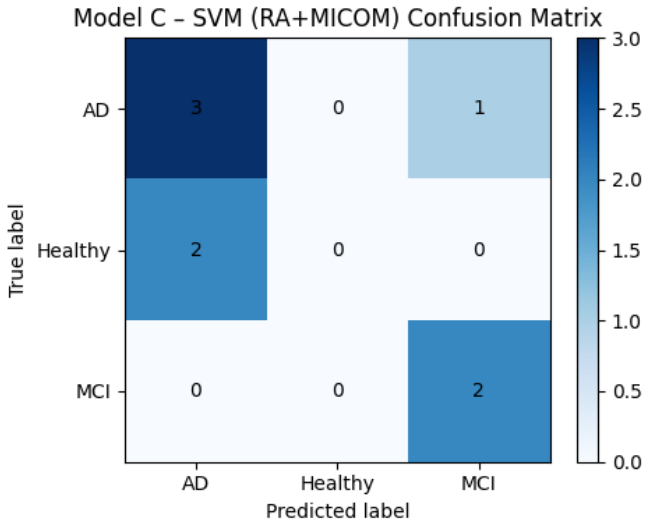

Figure 4 Confusion matrix for Model C (SVM, RA + MICOM). Predicted versus true class labels for Alzheimer's disease (AD), mild cognitive impairment (MCI), and healthy controls using combined taxonomic and metabolic features.

The addition of MICOM metabolic features results in a marked improvement in classification performance compared to the RA-only models. Test-set accuracy increased to 62% nearly doubling the performance achieved by both Model A (36%) and Model B (32%).

Table 5 Classification metrics for Model C (SVM, RA + MICOM). Class-specific precision, recall, and F1-scores calculated on the held-out test dataset. Weighted averages summarize overall model performance.

	Precision	**Recall**	**F1-score**
AD	0.60	0.75	0.67
Healthy	0.00	0.00	0.00
MCI	0.67	1.00	0.80
Weighted Average	0.47	0.62	0.53

Figure 4 presents the confusion matrix for Model C, and class-level metrics are summarized in Table 5. Prediction performance was strongest for MCI, with perfect recall (100%) and high precision (67%) yielding an F1-score of 0.80, indicating excellent separation of this transitional disease state once metabolic features were introduced. Classification performance for AD also improved substantially achieving 75% recall and an F1-score of 0.67.

In contrast, classification of healthy controls remained poor, with 0% precision and recall. As shown in the confusion matrix, healthy samples were consistently misclassified as disease states.

Weighted average metrics (precision = 0.47, recall = 0.62, F1 = 0.53) align with the strong MCI and AD classification performance while reflecting reduced performance for the healthy group.

3.6 Machine Learning Models: Model D

A Random Forest (RF) classifier was trained using the combined feature set including both relative abundance (RA) taxonomic features and functional metabolic outputs derived from MICOM modeling. The ensemble consisted of 500 decision trees with balanced class weighting to compensate for minor class imbalance. Performance was evaluated on an independent held-out test set to assess real world generalization following incorporation of functional metabolic features.

Model D produced the strongest overall performance among all models evaluated, achieving a test-set accuracy of 62%, an improvement relative to RA-only models (Model A: 36%; Model B: 32%).

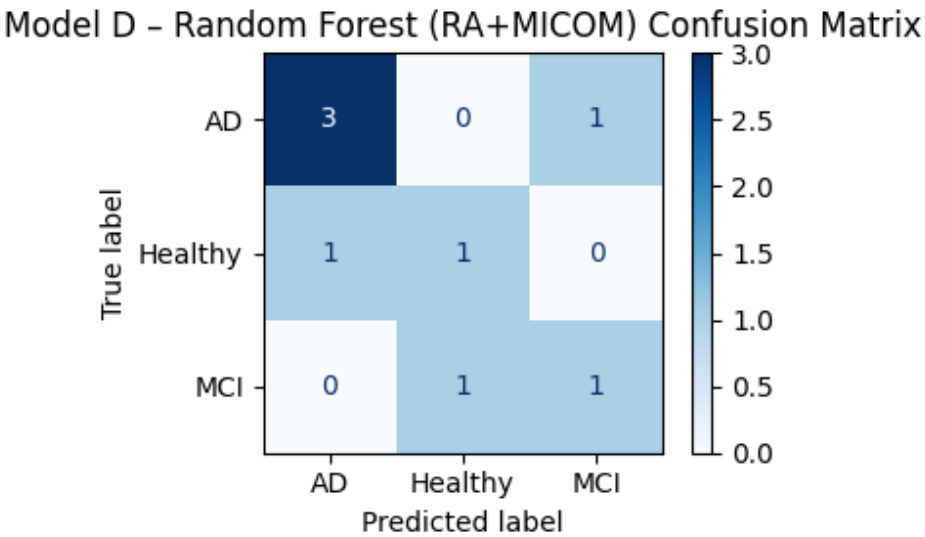

Figure 5 Confusion matrix for Model D (Random Forest, RA + MICOM). Predicted versus true class labels for Alzheimer's disease (AD), mild cognitive impairment (MCI), and healthy controls using combined taxonomic and metabolic features.

Table 6 Classification metrics for Model D (Random Forest, RA + MICOM). Class-specific precision, recall, and F1-scores calculated on the held-out test dataset. Weighted averages summarize overall performance.

	Precision	**Recall**	**F1-score**
AD	0.75	0.75	0.75
Healthy	0.50	0.50	0.50
MCI	0.50	0.50	0.50
Weighted Average	0.62	0.62	0.62

Figure 5 presents the confusion matrix for Model D, and class-level performance metrics are summarized in Table 6. The model exhibited robust classification of Alzheimer's disease samples achieving 75% precision, 75% recall, and an F1-score of 0.75. Predictions for the MCI group were moderately improved (precision = 50%, recall = 50%, F1 = 0.50). Healthy controls also demonstrated moderate classification performance (precision = 50%, recall = 50%, F1 = 0.50).

Weighted performance metrics (precision = 0.62, recall = 0.62, F1 = 0.62) aligned closely with the overall accuracy and confirmed the improved balance of classification across all clinical categories compared to RA-only models. Notably Model D maintained

strong AD detection while avoiding the extreme class collapse seen for healthy samples in Model C.

3.7 Machine Learning Models: Comparison

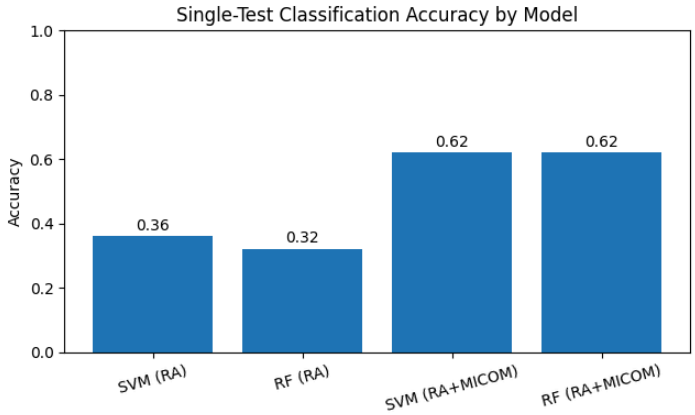

Figure 6 Single-test classification accuracy across all models. Comparison of held-out test accuracy for RA-only and MICOM-enhanced classifiers.

Table 7 Final weighted-average performance metrics for all evaluated models. Weighted precision, recall, and F1-scores summarizing aggregate classification performance.

Model	Accuracy	Precision	Recall	F1-score
SVM (RA)	0.36	0.36	0.36	0.36
RF (RA)	0.32	0.33	0.32	0.32
SVM (RA+MICOM)	0.62	0.47	0.62	0.53
RF (RA+MICOM)	0.62	0.62	0.62	0.62

The two RA-only models, SVM (RA) and Random Forest (RA), achieved low test accuracies of 36% and 32%, respectively. Correspondingly low weighted precision, recall, and F1-scores (all approximately 0.32–0.36) indicate limited discriminative capacity when relying exclusively on microbial taxonomic composition. Although both models obtained higher mean accuracies during cross-validation (~50–60%), this did not translate to independent testing, reflecting substantial overfitting and poor generalization. These findings confirm that taxonomic relative abundance data alone lack sufficient resolution for robust multiclass classification of cognitive disease states. In contrast, models incorporating functional metabolic features derived from MICOM exhibited markedly improved test performance. Both SVM (RA+MICOM) and Random Forest (RA+MICOM) reached a test accuracy of 62%, representing a near doubling of

classification performance to the RA-only approaches. The incorporation of metabolic flux information substantially increases weighted recall (0.62 for both models) and overall F1-scores (0.53 for SVM and 0.62 for Random Forest), demonstrating enhanced sensitivity and class-level balance.

Between the two MICOM-enhanced classifiers, Random Forest (RA+MICOM) emerged as the most stable and balanced model. Whereas SVM (RA+MICOM) maintained higher precision and recall for AD and MCI but failed to identify healthy controls, the Random Forest demonstrated more evenly distributed performance across disease categories reflected by matched weighted precision, recall and F1-scores of 0.62. This consistency indicates improved class separation and reduced bias toward diseased groups compared to the SVM alternative.

Overall, the comparative analysis underscores that feature informativeness rather than model choice was the dominant driver of classification success. The dramatic improvement following MICOM feature integration highlights the importance of functional metabolic modeling for diagnosing AD beyond microbial taxonomy. Among all evaluations Model 4 - Random Forest (RA+MICOM) achieved the strongest overall performance profile, supporting its designation as the optimal classifier within the study.

4. DISCUSSION

4.1 Relative Abundance and Principal Component Analysis

Relative abundance of genera remained fairly consistent across AD, MCI, and healthy subsets. A few genera, listed in Table 1, had notable differences in mean relative abundance (MRA) across conditions. However, even *Escherichia*, the genus with the greatest inter-condition range (ICR ≈ 14%), exhibited only modest differences in MRA. This demonstrates that variation in MRA remains minimal across condition subsets and therefore has limited potential to serve as a standalone feature for classifying disease state.

Furthermore, although a few genera distinct to individual conditions were identified, they appeared at low relative abundance. These genera may represent sample-specific variation rather than disease-associated patterns. Due to their limited presence in community composition, they are not suitable determinants of condition and were excluded as features.

Principal component analysis of prevalence filtered datasets (10% and 20%) showed extensive overlap of scores across AD, MCI, and healthy samples. Increasing the prevalence threshold did not significantly enhance separation, indicating that low-prevalence genera are not responsible for lack of structure between groups. This

implies that the principal axes of variation in genus-level relative abundance are unable to separate the samples into distinct condition groups. The first three principal components, for both prevalence thresholds, capture only 20-25% of the total variance. This low explained variance is consistent with the high-dimensionality and limited multivariate structure of the relative abundance datasets. Future analyses with species-level taxonomic resolution may elucidate stronger clustering of condition groups.

4.2 MICOM Metabolic Flux and Community Growth Analysis

As aforementioned, the MICOM feature set was reduced in size from 93 samples to 25 successfully reconstructed communities due to missing GEM models. The limited dataset introduces uncertainty about whether the reconstructed communities truly capture biologically relevant and metabolically meaningful differences across conditions. The small sample size leads to increased susceptibility to noise and offers poor generalization of observed metabolic patterns to unseen samples.

Despite these limitations, the reconstructed communities did suggest disease-specific metabolic activity across AD, MCI, and healthy microbiomes. Metabolic reaction fluxes demonstrated large inter-condition differences across several exchange reactions. AD and MCI communities had decreased secretion of xyluglucan, a plant cell-wall polysaccharide [15], as compared to healthy samples. This pattern suggests diminished microbial processing of dietary fiber components. Additionally, AD and MCI communities displayed decreased uptake of glucose, indicating reduced microbial carbohydrate metabolism. MCI communities had lower uptake of acetate and carbon dioxide, suggesting disturbances in short-chain fatty acid (SCFA) metabolism and reduced metabolic activity [16]. Lastly, AD communities demonstrated a substantial increase in 7-oxocholate secretion, indicating microbial bile acid (BA) dysregulation [16].

These disruptions in SCFA and BA metabolism in MCI and AD groups, respectively, are consistent with recent findings reporting altered SCFA and BA levels in neuro-degenerative diseases [16]. Overall, the significant differences in metabolic activity between diseased and healthy groups reflect dysbiosis in MCI and AD samples. Community growth rates were fairly similar across all conditions, suggesting that community-level production is not a strong indicator of disease state. However, AD samples did experience slightly elevated growth rate than MCI and healthy samples. This indicates that overall metabolic activity is comparable across groups, even though specific metabolic pathways vary substantially.

4.3 Model A and C Comparison: Support Vector Machines

The support vector machine classifier trained solely on RA features (Model A) achieved a 5-fold cross-validation (CV) accuracy of ~58.5%, indicating moderately strong performance on training data. However, this performance did not translate to independent testing, where accuracy decreased to 36%, demonstrating poor generalization. While precision, recall, and F1-scores were low across all conditions, they were especially poor for healthy samples. This indicates that while Model A could occasionally distinguish between AD and MCI, it struggled to correctly identify healthy samples.

The discrepancy between cross-validated accuracy and single-test performance reflects overfitting and reinforces the limited ability of genus-level RA features to capture functional disease signatures necessary for reliable diagnosis. These limitations align with the minimal MRA-ICR differences and diffused multivariate structure observed in PCA and highlight the importance of incorporating functional metabolic features.
In contrast, the support vector machine classifier trained on both RA and MICOM features (Model C) achieved a substantially improved test accuracy of 62%. It demonstrated a ~20% increase in precision, recall, and F1-scores for AD and MCI as compared to Model A, suggesting that MICOM-derived flux measurements capture disease-relevant metabolic signatures associated with cognitive impairment.

However, Model C's performance for healthy samples was drastically reduced to 0% across all metrics. Although both SVM models faced difficulty distinguishing healthy microbiome patterns, addition of MICOM features reduced dataset size, limiting capability of Model C to learn from the healthy subset. Only five healthy samples were available for training and two for testing in the stratified 70-30 train-test split. This made it challenging for Model C to generalize to the testing set for the healthy samples, likely compounded by overlapping metabolic profiles between healthy and diseased groups. Overall, MICOM-derived flux outputs substantially enhance model discriminative capability by capturing biologically meaningful microbial metabolic activity associated with AD progression. The improved performance of Model C justifies the incorporation of metabolic features to overcome the weak class separability provided by taxonomic features alone.

4.4 Model B and D Comparison: Random Forest

The Random Forest classifier trained solely on RA features (Model B) achieved a similar 5-fold cross-validation (49%) and testing accuracy (32%) to that of Model A, confirming weak generalization and marginal improvement over chance-level performance in a three-class setting. Similar to the SVM model, Model B exhibited low precision, recall, and F1-scores across all conditions. However, it achieved slightly

better performance (~50% for all metrics) in classifying MCI samples as compared to Model A (~40%). This suggests that transitional disease states (MCI) exhibit the most detectable microbial taxonomic signatures. This trend is reflected in the confusion matrix (Figure 3), where correct classifications are concentrated primarily within the MCI category.

Despite the greater modeling flexibility offered by ensemble methods, Random Forest did not overcome the fundamental limitations of the RA feature set. While modest improvements were observed for MCI detection, genus-level RA features remain insufficient for balanced multiclass classification.

In contrast, the Random Forest classifier trained on both RA and MICOM features (Model D) achieved the strongest overall performance across all four models. It reached a test accuracy of 62% and had strong performance in predicting AD samples (75% across all metrics), indicating high sensitivity and specificity for AD when metabolic data were included. It matched the performance of the SVM-based Model C while demonstrating more balanced class behaviour. The improved performance as compared to Model B is consistent with the observed potential of MICOM features to discriminate between disease states.

Healthy and MCI samples were classified correctly 50% of the time, reflecting improvement over Models A-C, where healthy detection was consistently poor. However, the overall performance for healthy and MCI classification remains limited. This can be attributed to the reduced MICOM subset size (n = 7) for MCI and healthy conditions, which restricts Model D's ability to learn disease-specific patterns from the training data and impairs generalization to unseen data.

The confusion matrix (Figure 5) illustrates reduced off-diagonal error compared to earlier models, particularly for AD samples, which had fewer misclassifications into other disease categories. While limited sample size constrained stable separation between healthy and MCI groups, incorporation of metabolic features resulted in clearer class boundaries overall. In summary, Model D demonstrates that combining metabolic modelling with ensemble learning (RF) provides the most consistent and biologically meaningful classification performance in this study. The model's balanced sensitivity across AD, MCI, and healthy conditions underscores the importance of integrating functional metabolic activity with taxonomic composition for accurate microbiome-based cognitive disease prediction.

4.5 Limitations and Improvements

Model C and Model D classification performance was primarily constrained by limitations in the MICOM feature set. Metabolic modelling was unable to identify genome-scale metabolic models for many of the identified genera, reducing the available dataset from 93 to 25 samples. This substantial reduction exacerbated class imbalance, as the MICOM dataset contained fewer healthy and MCI samples relative to AD. These factors reduced stability of stratified train-test splits and introduced bias into the learned patterns due to limited representation of key metabolic features. Although reconstructing missing GEMs could improve sample representation and feature completeness, it would require extensive additional research to characterize metabolic networks and community level microbial fluxes.

Additional methodological limitations may have influenced performance across all models. The MICOM pipeline used a default Western dietary medium and a fixed cooperative trade-off parameter, constraining the ability of subsequent features to reflect biologically realistic metabolic and community interactions. This can be improved by defining a diet medium similar to that of the study participants who resided in Hangzhou, China. And cooperative-tradeoff parameters can be iteratively tuned to achieve better performance in downstream classification modelling. Increasing resolution of taxonomy to species-level may also improve relative abundance features and could reduce the weak multivariate structure observed in genus-level relative abundance PCA.

Furthermore, tuning model hyperparameters could be used to achieve optimal model performance. For SVM models, exploring different combinations of regularization strength (C), kernel parameters, and local influence of training samples (gamma), may yield configurations that maximize 5-fold cross-validation accuracy. Similarly, RF may be optimized by tuning hyperparameters such as number of decision trees (iterations), maximum tree depth, minimum samples per split, and feature selection methods (ex. sqrt vs. log2). These suggestions could enhance model stability and improve generalization to unseen samples.

5. CONCLUSION

This study has developed a model that uses computational techniques including machine learning to predict stages of AD using gut microbiomes. It combined relative abundance taxonomic data with metabolic flux profiles developed by MICOM to analyze the relative composition and function of the microbiomes from healthy, MCI, and AD participants as diagnostic tools.

5.1 Key Findings

Our analyses indicated that distinct changes occurred in both microbial composition and metabolic activity across each stage of AD. Escherichia, Bacteroides, and Blautia are

the predominant genera contributing to between-group differences based on relative abundance analyses. While principal component analysis indicated some degree of overlap between the groups, the overlap was such that there was limited discriminatory capability. It was determined through MICOM modeling that MICOM produced a better understanding of functional metabolism and additionally provided metabolic disruption with large flux differences for xyluglucan, glucose, acetate, CO_2, and the 7-oxocholate exchange, which appear to represent the most significant metabolic disruption associated with AD.

However, when these features were used for machine-learning classification, models trained solely on RA data showed overfitting and generally did not generalize well. These observations also underscore that taxonomic composition alone does not capture AD-related biological signals and motivates the need for metabolically informed features. While smaller sample availability precluded extensive evaluation of MICOM features, observed metabolic flux differences indicate strong potential to improve diagnostic accuracy when combined with RA features.

5.2 Evaluation

It demonstrated how the diagnostic challenge for Alzheimer's disease can be approached using microbiome data. It was able to successfully implement a reproducible engineering pipeline from 16S processing in QIIME2 to GEM-based metabolic modeling and machine learning classification, showing how engineering tools can bridge biological data with predictive analytics.

Importantly, this study shows that looking at metabolic activity rather than relying only on microbial abundance provides a clearer, more mechanistic understanding of the changes happening along the gut–brain axis. Although the RA-based SVM and Random Forest models show limited performance, the metabolic insight accrued from MICOM suggests that functional microbiome modeling may be indispensable in building accurate AD diagnostic systems. Collectively, these findings support more widespread efforts to create biological and non-invasive diagnostic tools that are capable of being used on a large-scale level to help identify neurodegenerative diseases.

5.3 Future Outlook

In general, the overall limitation of this research is the small sample size of MICOM, lack of GEM coverage for some taxa, and restricted interpretability influenced by taxonomic sparsity and dataset imbalance. Future work should expand the dataset in order to ensure fuller GEM representation and incorporate multi-omics layers like metabolomics or transcriptomics that will enhance the robustness of the model.

Future community-based metabolic reconstruction and use of ensemble machine-learning methods have the potential to provide less variability and, therefore, more accurate stage predictions than the current methods.

If this framework is developed with larger and more diverse cohorts of individuals using optimized integration of metabolic flux features, it may provide the foundation for a clinically useful, non-invasive diagnostic platform for early diagnosis of Alzheimer's disease. This is a trend in biomedical engineering; the conversion of complex biological signals into clinically useful diagnostic tools.

6. SUPPLEMENTAL MATERIAL

6.1 GitHub link: https://github.com/haniya-naeem/AD-Microbiome-ML-Classification

6.2 Dataset: Liu, P., Wu, L., Peng, G., Han, Y., Tang, R., Ge, J. et al. Altered microbiomes distinguish Alzheimer's disease from amnestic mild cognitive impairment and health in a Chinese cohort. Brain Behav Immun. 2019 August; 80: 633-643. doi: 10.1016/j.bbi.2019.05.008.

6.3 MICOM: Diener, Christian. et al. MICOM: Metagenome-scale modeling to infer metabolic interactions in the human gut microbiota. mSystems. 2020 January 21; 5(1): e00606-19. doi: 10.1128/mSystems.00606-19

6.4 SILVA: Robeson M. S. et al. RESCRIPt: Reproducible sequence taxonomy reference database management. PLos Comput Biol. 2021 November; 17(11): e1009581. doi: 10.1371/journal.pcbi.1009581

6.5 QIIME: Bolyen E, Rideout JR, Dillon MR, Bokulich NA, Abnet CC, Al-Ghalith GA, et al. Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2. Nat Biotechnol. 2019 August; 37(8): 852-857. doi: 10.1038/s41587-019-0209-9.

6.6 AGORA2: Heinken, A., Hertal J., Acharya, G. et al. Genome-scale metabolic reconstruction of 7,302 human microorganisms for personalized medicine. Nat Biotechnol. 2023 January; 41: 1320-1331. doi: 10.1038/s41587-022-01628-0

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Section *1.2 - Existing Solutions and Gaps is reproduced from the group's previously submitted project proposal for ENGG 680. This reuse was intentional to maintain continuity and accuracy of the original literature review and referenced background research developed for the project. Permission to do so was granted by the course instructor. All external sources cited remain properly attributed. The remaining sections of this final report including methodology implementation, analysis, results, discussion, and conclusions constitute original work prepared specifically for this submission.*

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